

KIMSIE PHAN

📍 IIT ISM Dhanbad ✉ phankimsie03@gmail.com ☎ 91 9289825725 🔗 kimsiephan.github.io
in PHAN KIMSIE 🌐 kimsiephan

Research Interest

- Federated Learning–Based Air Quality Anomaly Detection in Underground Mining Environments.
- Multimodal Human Activity Recognition Using Federated Learning.
- Integrating AI and Learner Psychology for Inclusive and Personalized Education.

Scholarships

- Jul 2024 - May 2026: **Master's Degree** from Indian Council for Cultural Relations (ICCR)
- Nov 2021 - Jun 2024: **Bachelor's Degree** from Indian Council for Cultural Relations (ICCR)

Work Experience

Mathematics Teacher <i>Tromung Chrum Secondary School</i>	<i>Kampong Chhnang, Cambodia Oct 2021 - Present</i>
Mathematics Lecturer <i>National University of CheaSim Kamchaymear (NUCK)</i>	<i>Prey Veng, Cambodia June 2024 - July 2024</i>
Mathematics Teacher <i>Samdach Akak Moha Thoma Pouthisal Chea Sim Prek Anhchan High School (New Generation School) - NGS PA</i>	<i>Kamdal, Cambodia Oct 2020 - Oct 2021</i>

Teaching Experience

- Deep Learning; for UG, Mathematical Association of Cambodia (MAC), Oct 2025 - Present
- Machine Learning; for UG, MAC, Dem 2024 - Sep 2025
- Mathematics Analysis II; for UG, NUCK, Jun 2024 - Jul 2024
- Mathematics; for Grade 7; NGS PA, Oct 2020 - Oct 2021

Education

Indian Institute of Technology (Indian School of Mines), Dhanbad <i>MSc in Mathematics and Computing</i> ◦ CGPA: 7.3/10 ◦ Courses: Probability, statistics inference, abstract algebra, differential equations, numerical method, data structure, data mining, topology, functional analysis, operation system, database management system, research methodology, computational fluid dynamic, design algorithm and analysis, and optimization techniques. ◦ Project: Federated Learning for Anomaly Detection in Mining Sensor Networks ◦ Advisor: Dr. Tamoghna Ojha	<i>July 2024 - Present</i>
University of Delhi <i>BSc (Hons.) in Mathematics</i> ◦ CGPA: 8.041/10 ◦ Courses: Python, calculus, algebra, real analysis, probability, computer networks, latex and html, differential equations, linear algebra-I & II, computer algebra, numerical analysis, discrete math, complex analysis, abstract algebra, mathematical finance, and linear programming. ◦ Project: DYNAMICS OF THE LORENZ CURVE ◦ Advisor: Mr. Vishal Dhawan	<i>Nov 2021 - June 2024</i>

Regional Teacher Training Centre, Kandal*Nov 2018 - Aug 2020**Teaching Training Diploma*

- **Percentage:** 89%
- **Courses:** Mathematics, pedagogy, teaching methodology, logic, social sciences, physic, history

Pannasastra University of Cambodia*Apr 2019 - Jul 2020**Intensive English for Academic Purposes (IEAP) Diploma*

- Percentage: 83.91%
- **Courses:** Intensive English for Academic Purposes (IEAP)

Bun Rany Hun Sen High School, R'lea Bier*Oct 2015 - Aug 2018**High School Diploma*

- Percentage: 85%
- **Courses:** Mathematics, physics, chemistry, biology, Khmer literature, English, history, geography

Invited Talks

MAC, Online Seminar Title: Introduction to Federated Learning [Slide ↗]	<i>29 May 2025</i>
MAC, Online Seminar Title: Gradient Descent Algorithm [Slide ↗]	<i>29 Sep 2024</i>
MAC, Online Seminar Title: Three Basic Epidemiological Models [Slide ↗]	<i>22 Feb 2023</i>
MAC, Online Seminar Title: How to Solve System of Linear Equation Using Matrix Notation [Slide ↗]	<i>22 Aug 2021</i>
MAC, Online Seminar Title: Quotient Groups and Fundamental Theorem of Homomorphisms [Slide ↗]	<i>15 May 2021</i>

Membership

- Cosine Lab of IIT ISM Dhanbad
- Mathematical Association of Cambodia (MAC)

Professional Service

- Peer Reviewer, Scientific Reports (Springer Nature) - 2025 Reviewed four manuscripts in the field of artificial intelligence.
- Review Activity for IGI Global Book Chapter Peer Reviews (2).

Languages

Technologies: PyTorch, Latex, Python, Matlab, R, C++, HTML & CSS, SQL, Mathematica

Communication: Khmer (mother tongue), English (fluent)

Reference

Prof. Tamoghna Ojha (Thesis supervisor)

Assistant Professor, Department of Mathematics and Computing

Indian Institute of Technology (Indian School of Mines), Dhanbad, India

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